Radar de Señal Temprana: Master Plan for the 30X "Radar de contenido" Hackathon

Version: Saturday, Oct 17, 2026, 18:05; **revised 18:35** (Bogotá, UTC-5) for the ~$250 Google Cloud credit (GCAM in scope), GPU graph visualization, graph DB choice and the gated TGNN plan (details in `graph_viz_tgnn_gcam.md`). **Deadline:** Sun Oct 18, 12:00 (UTC-5).

Status of this document: this is the single source of truth. It consolidates `report.md`, `gdelt_graph_analysis.md`, `bq_free_tier_methodology.md` and `graph_viz_tgnn_gcam.md` (see References) plus decisions made since.

- **(measured)** = we ran it.
- **(verified)** = read on a primary source today.
- **(user-confirmed)** = seen by the user in their BigQuery console.
- **(estimate)** = our projection.
- **UNVERIFIED** = not yet checked.

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0. One-paragraph summary

We build an **Early Signal Radar** that tells 30X **which AI topics are about to land on LinkedIn, why, and from which emotional angle to tell them** to LatAm executives. LinkedIn bans automated collection, so we measure the open upstream chain instead:

1. Papers, models and repos (arXiv, Hugging Face, GitHub)
2. Tech-social (Hacker News, X counts, Bluesky)
3. Global news (GDELT, 65 languages, with tone)
4. Reference attention (Wikipedia es/pt/en)
5. LinkedIn, observed only through 30X's own analytics and a community sensor

The backbone is a **daily, normalized, weekday-adjusted GDELT time series on BigQuery's free tier**. It carries **tone, extremity, polarity and controversy as first-class signals**, plus **GCAM emotions (arousal, fear, hope, uncertainty; native Spanish and Portuguese dictionaries) over 2 years for ~$10–15 of the Google Cloud credit**, and weekly co-occurrence graphs (igraph/cuGraph, NPMI, Leiden). The graph is shown in a **two-tier GPU visualization**: a full 90-day temporal graph and a pruned, explainable story view.

Six scores combine into the weekly top 10. Each top-10 topic ships with a Spanish adaptation brief. Evidence for the judges is a backtest (lead time, precision@10) and measured filter recall.

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1. The brief and its constraints

Challenge (Reto 30X "Radar de contenido"): discover, **reliably and without spending a fortune**, which AI topics and formats are taking off on LinkedIn, Instagram, TikTok and YouTube. Explain **why they work**, whether a trend is **just starting or already copied by everyone**, and whether it fits **LatAm executives and teams learning AI**.

Prototype bar: answer questions like *"top 10 AI topics growing this week on LinkedIn, why, and how to adapt them."*

Hard constraints (from the brief):

- Public content only.
- No personal data.
- No techniques that violate terms of service.
- No content behind third-party logins.

What the judges reward (adversarial reading, `report.md` §2):

Requirement	How we meet it
Reliability	Backtest plus measured precision/recall
Low cost	Lean tier ≈ US$40–55/month (estimate) and a $0 BigQuery month
Topics *and* formats	Topic ranking plus a format/angle layer
Explanation, not just ranking	Every rank carries its reasons
Lifecycle stage	Stage and saturation score

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2. Concept, alternatives and final scope

2.1 Flagship: Early Signal Radar

- A multi-agent pipeline: scouts → normalizer → topic miner → signal analyst → diffusion/saturation → audience-fit judge (Jev) → explainer → ranker → Spanish adaptation writer → weekly radar with editor feedback.
- **Pitch line:** *"We tell you what's about to land on LinkedIn, and from which emotional angle to tell it."*

2.2 Alternatives considered (`report.md` §5)

Path	Idea	Strength	Why not the core
A. Hook Lab	Multimodal "why it worked" on YouTube Shorts (Gemini on public URLs), outperformance vs. the creator baseline, SHAP	Great visuals; the "why" criterion	Formats only; slow to label at scale. **Adopted as a slice:** 20–40 videos for the sentiment validation and format chips
B. Community Sensor	WhatsApp/Slack bot: members voluntarily share LinkedIn/IG posts from their own feeds	**The only clean LinkedIn path**	Needs a community to show volume. **Adopted as a demo feature** (mock + 1 live flow)
C. Lead-Lag Forecaster	News→social, EN→ES lags; Chronos-2 / TimesFM	The "early" criterion	**Merged into the flagship** as the Stage score and backtest
D. 30X Voice Model	SFT/DPO a small model as the writer	Ambitious	Too risky in 18 h. **Rejected for the hackathon**; on the roadmap

2.3 Final recommended scope (what we ship by 12:00)

1. **GDELT daily topic radar on BigQuery** (`radar.daily_topic` + rolling view), 2 years where the backfill completes. **Minimum: the last 6 months.**

2. **Six scores** and a **weekly top 10** (Spanish UI) with "why" bullets, an emotional framing angle per topic, and an adaptation brief.

3. **Two-tier graph visualization** (§11.1):

    - Tier 1 "wow": the full 90-day temporal graph on **Graphistry Hub** (server-GPU layout, timebar, iframe). Backup: **cosmos.gl** with cuGraph-precomputed coordinates.
    - Tier 2 "story": a topic ego-graph replay in **sigma.js** with evidence on click.
    - Plus a timeline showing HN → GDELT → Wikipedia onsets.

4. **Backtest card** (lead time, precision@10 vs. baselines) and **filter quality card** (precision on the gold set, recall on the control sample).

5. **Community sensor** demo flow and the **X counts** cross-check for the top 10 (paid per request, a few dollars).
6. **GCAM emotions**: the last ~108 days first (for the demo), then the full 2 years overnight.
7. Stretch:
    - **TGNN**: TGM library; TGN/GraphMixer vs. EdgeBank/PopTrack. Gated by the go/no-go in §11.3.
    - Memgraph "ask the graph" moment
    - Kepler.gl LatAm geo arcs
    - gsg_docembed recall expansion

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3. Sources and terms-of-service verdicts

Source	Use	ToS verdict
LinkedIn	Only 30X's own Page analytics plus **voluntary community submissions**	User Agreement (Nov 3, 2025) bans "crawlers, browser plugins and add-ons" (verified, `report.md` §7). **No scraping, no extension.** hiQ v. LinkedIn ended in a consent judgment against hiQ
Community sensor (Slack/WhatsApp bot)	The clean LinkedIn/IG/TikTok signal: links people choose to share	Clean when there is a privacy notice and consent (Ley 1581 / Decreto 1377); we store topic-level aggregates only
X API v2 (pay-per-use)	Counts for the top-N topics (**$0.005/request**, verified); a few exemplar reads	Clean (Developer Agreement)
Hacker News (Algolia)	Early tech signal; full history for backtests	Clean, free
Wikipedia pageviews	es/pt/en attention; "Spanish article exists" mainstreaming marker	Clean; send a descriptive User-Agent
YouTube Data API + RSS + Gemini on public URLs	Format exemplars, outperformance vs. the creator baseline, hook features	Clean within quota; refresh/delete non-authorized data within 30 days
Bluesky Jetstream	Free firehose; creator diffusion	Clean (open protocol; honor deletions)
Hugging Face / arXiv / GitHub	Supply-side lead (models, papers, repos)	Clean (APIs; arXiv 1 request per 3 s)
GDELT (BigQuery + raw 15-minute files)	News backbone, tone, entities, LatAm coverage	Open data with attribution

Source	Use	ToS verdict
Instagram Graph API	Business Discovery for reference creators	Needs App Review. **Post-hackathon**
TikTok Research API / Meta Content Library	—	Academic only. **Not usable**
TikTok Creative Center	Manual editor check	**No scraping**
Reddit, Product Hunt	—	Commercial-use approval needed. **Skip**
Apify / Bright Data scrapers	—	**Excluded** (violates the brief's "no ToS-violating techniques")

Honest framing: we present "LinkedIn growth" as **predicted LinkedIn uptake** plus **community-observed LinkedIn signal**, and we say so explicitly.

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4. Jev: where it fits

- **What Jev is.** TypeSafe's calibrated decision model, `typesafe/jev-1.13` via the OpenRouter Decisions API.
  - It returns typed answers (`noul`, `choice`, `score`) with probabilities and generates no text.
  - Vendor-stated price: $0.042/M input tokens, output free.
  - 30X ran a "Demo con Jev" session for this hackathon (verified, `report.md` §3).
- **Use it for** topic-level and item-level decisions:
  1. Is the item AI? (`noul`, a precision pass on ambiguous rows such as the Italian "ai" or "Claude" as a person)
  2. Audience fit for LatAm executives, 0–4 (`score`), which **is** the Audience-fit score
  3. Lifecycle stage (`choice`)
  4. Framing angle (`choice` over the angle set in §10.5)
  5. Brand-safety (`noul`)
- **Pattern:** *Jev decides, the LLM explains, code ranks.* UNVERIFIED: Jev rate limits and latency at our volume.

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5. GDELT strategy (BigQuery-first)

5.1 User-confirmed facts (Oct 17, 2026, user's console, "My First Project", on-demand billing)

- **`gdelt-bq.gdeltv2.gkg_partitioned`:** ~1.84B rows, ~20.5 TB. Last modified 2026-10-17 22:33 UTC (17:33 UTC-5). Newest DATE 20261017224500, so it is **live within ~15 minutes**.
- **Other live tables:**

Table	Size
webngrams	133 TB

Table	Size
gfg_partitioned	80 TB
events	370 GB
eventmentions	557 GB
gsg_docembed	3.6 TB / 893M rows
geg_gcnlapi	—
gsg_iatvsentembed	—
web_pos / web_2grams / web_1grams	—

- **Stale, skip:** cloudvision (2022), iaradio_* (2023), iatv_* (2024).
- **A 2-day test scan** (3.64 MB) returned **477,517 rows**, against ~645k expected from the raw files (≈74%). **Open item:** Q1 reconciles it. Until then we normalize by BigQuery's own totals (`radar.daily_totals`), which is internally consistent.

5.2 What we take and what we skip

- **Take:** `gkg_partitioned` via one **single-pass monthly INSERT** of the graph-ready columns into our own `radar.gkg_ai`. Columns:
  - gkg_id, ts, source, url, title, lang
  - **V2Tone, all 7 fields**
  - orgs, persons, themes, locations, allnames
  - Flags, plus a 0.2% control sample
- **Skip:**
  - (GCAM is now **taken**, via S7, into a narrow 46-dimension table; see §10.6)
  - `webngrams` via SQL (use raw minute files if needed)
  - gfg, events/mentions, TV/radio/cloudvision
  - The unpartitioned `gkg`
- **Measured raw facts** (`gdelt_graph_analysis.md`):
  - 2 years = 136,959 files, 937 GB compressed.
  - **Gap 2025-06-14 17:45 → 2025-07-02 02:15 UTC** (417 h).
  - English monthly volume is falling ~22%, so we **always model shares**.

5.3 `gsg_docembed` and `geg_gcnlapi` verdicts

Table	Verdict	Use	Never
`gsg_docembed` (USEv4, 512-d, ~4 KB/row)	**7/10, stretch**	Recall expander and clustering space, pulled via the **Storage Read API** (300 TiB/month free reads; ~$1.7 egress for AI rows, estimate)	`SELECT docembed` over long windows (~1 TiB for 2 years)
`geg_gcnlapi` (Cloud NL entities, ~100k-article daily sample, MIDs + salience)	**8/10 for entity resolution** (MID → Wikidata P646/P2671)	Entity resolution; size unknown until the Q4 dry run	Volume series (it is a sample): 4/10

6. Free-tier methodology and cost ledger

Mechanics (verified, `bq_free_tier_methodology.md` §1):

- On-demand pricing is $6.25/TiB with **1 TiB/month free**.
- You are billed per **column referenced** across the pruned partitions. Row filters, LIMIT and joins don't reduce bytes.
- 10 MiB minimum per query.
- Preview and `tabledata.list` are free. Extract jobs are free.
- The Storage Read API has 300 TiB/month free.

Per-column bytes (measured, B/row): GCAM 9,172 · V2Themes 1,133 · Themes 520 · Extras 433 · V2Locations 396 · AllNames 233 · URL 108 · **V2Tone 97** · V2Organizations 70 · V2Persons 43.

The graph-ready set comes to **2,575 B/row**, or **~0.57 TiB for 2 years** at the raw-file rate (×0.74 if BigQuery holds 245k rows/day).

V2Tone typed columns cost nothing extra. V2Tone was already part of the 2,575 B/row set; we now parse all 7 fields instead of 1.

Step	Bytes (estimate)	$
Guardrails: max bytes billed 50 GiB/query, daily quota 0.7 TiB, $1 budget alert	0	0
Q1 reconcile / Q2 column budget / Q3 7-day pilot ×6	~25 MB / ~1 GB / ~15 GB	0
S3 `daily_totals`, 2 years	~6 GB	0
S2 × 24 monthly slices (graph-ready + V2Tone)	**0.42–0.61 TiB**	0
S4 aggregates incl. `article_topic` + `daily_topic` rebuild (own tables)	~30–60 GB (+ ~3 GB per daily_topic rebuild)	0
S8 rolling view queries	<1 GB each	0
S6 daily script (MERGE + totals + 5-day topic rebuild)	~0.055 TiB/month	0
S7 GCAM, ~108 days for the demo (cap raised to 150 GiB per monthly job)	**0.22–0.30 TiB**	**~$1.4–1.9** (credit)
S7 GCAM, remaining 20 months (overnight)	**1.3–1.9 TiB**	**~$8–12** (credit)
S7d daily GCAM / S9 / S10 own-table rebuilds	~6 GB/day; a few GB per rebuild	~$1–2
Month-1 total without GCAM	**≈ 0.55–0.75 TiB**	**$0** (worst case $2–5)
Month-1 total with 2-year GCAM	**≈ 2.1–3.0 TiB**	**≈ $12–20** on the ~$250 credit

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7. Hybrid freshness

- **Daily:** S6, a scheduled script at 06:00 UTC (01:00 UTC-5):
    1. MERGE the last 2 ingestion partitions into `gkg_ai`.
    2. Refresh `daily_totals`.
    3. Rebuild 5 days of `article_topic` / `daily_topic`. - Scheduled queries need the Data Transfer API, so they are not available in the sandbox. UNVERIFIED: whether a

scheduled query accepts the multi-statement script; if not, split it into 4 schedules.

- **Intraday:** raw GKG 15-minute files over **https** (http hangs, measured), via `lastupdate.txt` and the PoC ingester, at 0 BigQuery cost. They feed the "last ingested N min ago" badge and the current partial day. Partial days are **flagged** and never ranked as complete days.
- **Rolling 7-day metrics are refreshed daily. Graphs are built weekly** (Monday-start weeks).

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8. AI taxonomy, filter quality and recall

Wide net in BigQuery, precision locally. Recall is free in BigQuery because bytes depend on columns, not rows.

Flags:

- `f_url`: multilingual URL slugs
- `f_url_short`: `/ai/`, `/ia/` with a language guard
- `f_title_strong` / `f_title_weak`
- `f_org`: AI lab organizations
- `f_ict`: World Bank ICT themes, only 41% of AI rows carry them (measured)
- `f_latam_loc` / `f_latam_src`
- **`f_control`**: a 0.2% FARM_FINGERPRINT random sample

Core AI = `f_url OR f_url_short OR f_title_strong OR f_org OR (f_title_weak AND f_ict)`.

Subtopics (`radar.topic_rules`, regex v1 on title + URL) take 11 values:

- agents
- llms_models
- chips_infra
- regulation_policy
- education
- jobs_work
- business_adoption
- safety_risk
- creative_media
- health_science
- funding_markets

Later, LLM labels replace the regex. They come from Nebius Qwen3-30B in batches of 20, with an embedding classifier routing only uncertain rows, and are loaded back with a free load job.

Measured baseline: v1 lexicon selectivity 1.41%, ~80% precision (single-annotator eyeball).

Recall measurement:

1. Label the AI-positive rows in the control sample.
2. Recall = caught by the net / all AI-positive control rows.
3. Precision comes from 300 stratified candidates, with Wilson CIs per language.

Targets: P ≥ 0.9 after the LLM pass, recall ≥ 0.85 (to be tested).

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9. Daily grain: normalization, seasonality, gaps

- **Grain:** `radar.daily_topic` = day × topic (`ai_all` + 11 subtopics) × lang_bucket (eng/spa/por/other) × latam flag.
- **Normalization:** `share_of_day` = n_articles / `daily_totals.n_all` for that day and language stream.
    - LatAm rows also use the stream denominator; there is no LatAm total.
    - Raw counts are **never** ranked, because GDELT volume declines ~22%/year.
- **Weekday seasonality:**
    - `z_dow`: today's share vs. the same weekday at lags 7/14/21/28 (needs ≥3 valid).
    - `z_7d`: rolling 7-day share vs. its distribution over days 34..7 back.
- **Growth:**
    - `growth_7d_log` = ln(share_7d / share_7d a week ago).
    - `accel_7d` = growth minus growth a week ago.
- **Gaps:**
    - The 2025-06-14..2025-07-02 outage and any day below 50% of the trailing 28-day mean volume are `is_missing` and come through as **NULL, never 0**.
    - A day with source data but no matches is a real 0.
    - 7-day sums need ≥5 valid days and are scaled to 7.
    - Backtests skip windows that overlap the gap.

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10. Sentiment as a first-class signal

10.1 Thesis

The user's thesis is that **both ends of the sentiment spectrum drive appealing content**. Fear (jobs, safety, deepfakes) and awe or optimism (breakthroughs, productivity) both outperform lukewarm coverage. So we track **intensity and divisiveness**, not just mean tone.

10.2 Measured calibration

Measured on 6,976 AI rows, Oct 9–16, 2026, from the PoC:

- **Tone** (V2Tone field 1): P10 −3.82 · P50 0.0 · P90 3.56 · SD 3.08.
- **Polarity**: P10 3.64 · P50 6.46 · P90 10.53.
- Median word count: 447.
- With **X = 3** (≈1 SD): 14.2% of rows > +3 and 15.6% < −3.

10.3 Signals

Per day in `daily_topic` and per rolling 7 days in `daily_topic_rolling`:

Signal	Definition	Reads as
mean_tone, sd, p10, p90	V2Tone field 1	Direction and spread
extremity	ABS(mean tone)	One-sided emotional coverage
mean_polarity	V2Tone field 4 (emotional word density, either sign)	Charge even when the average is neutral
positive_share / negative_share	Share of articles with tone > +3 / < −3	Size of each tail
controversy_index	`LEAST(1, polarity/10.5) × GREATEST(0, 1 − ABS(tone)/3) × 2·LEAST(pos_share, neg_share)`	Charged, neutral on average, both tails populated: a **divisive** topic. Range 0..1, baseline ≈ 0.2–0.3 on all AI coverage (estimate from the measured shares). Read it as a z-score vs. its own trailing window
tone_shift_7d	mean_tone_7d − mean tone of the 28 days ending a week ago	**Early signal:** a sentiment swing often precedes a volume burst (hypothesis, to be backtested)

10.4 Sixth score: Emotional charge

`EmotionalCharge = mean(z(extremity_7d), z(mean_polarity_7d), z(controversy_7d), |z(tone_shift_7d)|, z(arousal_rel_7d), z_fear_and_hope_7d)`. The last two terms come from GCAM (`radar.gcam_topic_rolling`): ANEW arousal relative to the same-day all-news baseline, and fear and hope rising together, with z computed per topic × language vs. its trailing 28 days. The equal weights are a placeholder, to be tuned on the validation test in §10.7.

10.5 Framing angle per topic

The framing angle feeds the Spanish adaptation brief. Rule, then a Jev `choice` check on the topic summary:

- **"Alerta / riesgo"** (fear frame): negative tail dominant, tone < −1.5.
- **"Oportunidad / cómo aprovecharlo"**: positive tail dominant, tone > +1.5.
- **"Debate"** (two sides; ideal for LinkedIn polls and carousels): controversy z > 1.5.
- **"Giro"** (story of change): |tone_shift_7d| z > 1.5.
- **"Explicador neutral"**: otherwise.

The brief includes the angle, two hooks in Spanish matching it, and evidence headlines from both tails.

10.6 GCAM (in scope with the ~$250 credit; S7 / S7d / S9)

- **Cost:** GCAM is billed in full for every touched partition, so cost depends on days, not rows. At the measured 9,172 B/row:

Window	TiB	$
30 days	0.06–0.08	$0.4–0.5
90 days	0.18–0.25	$1.2–1.6
2 years	**1.5–2.2**	**$9–14**

- **Plan:**

    - Run October, then September, August and July 2026 first (~108 days, ~$2). Then the remaining 20 months overnight.
    - Raise Maximum bytes billed to **150 GiB for the S7 job only**, then restore 50 GiB. Raise the daily quota temporarily, or spread the runs over 3 nights.
    - Add $25 and $50 budget alerts. Check that the credit covers BigQuery (UNVERIFIED).

- **Read once:** S7 parses GCAM for all rows of the month into a temp table. From it, it writes:

    - `radar.gcam_ai`: 46 dimensions for AI candidates plus the control sample;
    - `radar.gcam_daily_baseline`: the emotional climate of all news per day and language, at no extra GCAM cost.

- **Dimensions** (verified in the GCAM master codebook):

    - ANEW valence/arousal/dominance (v19.1–3)
    - LIWC sad/anger/anxiety/negemo/posemo (c5.31–35)
    - WordNet-Affect 1.1: alarm, anticipation, anxiety, awe, confidence, enthusiasm, hopefulness, insecurity, joy, negative-fear, positive-hope, security, surprise
    - Roget HOPE/FEAR (c9.896/898)
    - General Inquirer arousal/EMOT
    - Loughran–McDonald uncertainty/litigious (c6.6/c6.1)
    - Moral Foundations vices
    - Hedonometer, VADER, SentiWordNet, Lexicoder
    - **Spanish native:** Spanish ANEW v22.1/2, ML-SENTICON v23.17/18, Hedonometer v24.9
    - **Portuguese native:** c27.1/c27.2, Hedonometer v24.7

- **Measured coverage** (6,184 rows, Oct 17): ANEW and Hedonometer ~100%; **Spanish ANEW on 99% of Spanish-source rows; Portuguese c27 on 98% of Portuguese-source rows**. Native dictionaries run on the native text (verified, GKG 2.1 codebook: the `nwc` entry).

- **Proxies:** there is no "trust" dictionary in GCAM. Trust is proxied by confidence + security.
- **Topic level (S9):** fear, hope, anxiety, awe/surprise, anticipation, trust-proxy, uncertainty, anger and moral-outrage densities per 1,000 words, plus arousal relative to baseline. The rolling z-scores include `z_fear_and_hope_7d`.
- **Licensing:** GDELT open data (cite GDELT). The dictionaries carry their own citations and licences (LIWC is proprietary as a dictionary). We store GDELT's scores only. Redistribution of derived scores is UNVERIFIED.

10.7 Caveats and validation

- **News tone ≠ social reaction.** GDELT tone is a lexicon score of *journalists' text*, computed after machine translation for non-English text. It is not audience emotion.

- **Validation test** (before we claim it in the pitch):

    1. For 20–40 recent AI YouTube videos (Spanish and English) and the X posts matched to our top topics, compute **outperformance = views or engagement / the creator's median over their last 15 uploads** (YouTube RSS) or recent posts.
    2. Correlate with the topic's Emotional-charge score at publish time (Spearman), with a permutation p-value.

- The result goes on the evaluation card either way.

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11. Graph approach

- **Computation:** weekly rolling snapshots.
  - Nodes: org, person, theme, location, topic.
  - Edges: co-occurrence weighted by **NPMI** (`radar.edges_weekly` , n ≥ 3), backbone NPMI > 0.1.
  - Engines: igraph on CPU for the story view; **RAPIDS cuGraph on a Modal GPU** for the full graph (ForceAtlas2 layout warm-started week to week via `pos_list` , Leiden, PageRank).
- **Per-node metrics:**
  - Weighted degree and growth
  - New-neighbour rate vs. the prior 4 weeks
  - PageRank rank delta, k-core
  - Leiden communities matched across weeks (Jaccard ≥ 0.3) for birth/merge/split events
  - Kleinberg bursts
  - Tone, GCAM arousal, fear/hope per node-week (`radar.viz_nodes_weekly` , S10)
- **Measured PoC sizes:** 6,544 AI docs → 10,487 entities, 10,620 pairs with n ≥ 3.
- **Extrapolated 90-day graph (upper bound):** ≤ 0.33M nodes, ≤ 0.58M edges with n ≥ 3, ~7M co-mention events.

11.1 Two-tier GPU visualization (details: `graph_viz_tgnn_gcam.md` §2)

Tier 1, "wow" (whole 90-day temporal graph, ~0.1–0.33M nodes / 0.3–0.6M edges):

- **Primary: Graphistry Hub** (free tier).
  - Layout runs on **Graphistry's server GPUs**; rendering on WebGL2 in the browser.
  - Graphistry 2.53.0 opens 10M edges in 537 MB of browser memory (verified blog, Sep 3, 2026; "still choppy" at that size).
  - Timebar on the week/first-seen columns.
  - Embedded by iframe URL.
- **Backup: cosmos.gl** (MIT; v3, Jun 2026; ">1M nodes and links").
  - Renders **in the client GPU** from coordinates **precomputed by cuGraph on our Modal GPU**. cuGraph ran a 0.7M-vertex / 1.2M-edge ForceAtlas2 in 4.8 s on a V100, per RAPIDS.

Tier 2, "story" (explainable):

- **sigma.js v3 + graphology** (MIT), fed by S10's pruned weekly graph: ≤ 400 nodes and ≤ 2,500 edges per week; topic ego views ≤ 80 nodes.
- Interactions:

    - Week slider with tweened transitions
    - Node fill = tone or relative arousal; node ring = fear vs. hope
    - Edge tooltip with n, NPMI and first-seen week; new edges pulse
    - Leiden community colors with birth/merge badges
    - **Click a node** for evidence headlines (es/pt/en) with tone

- Backup: react-force-graph-2d.

Where the GPU runs, and the risk:

Where	Role	Risk
Graphistry cloud	Layout	Needs network; Hub resource limits
Client browser	Rendering in every tier	**Demo-laptop risk**: integrated GPU, hardware acceleration off
Our Modal GPU	Precompute only	None at demo time

Mitigations:

- Test on the actual demo laptop by 09:00 (Chrome hardware acceleration on).
- Cap cosmos.gl at 300k edges (150k on an integrated GPU).
- Keep a recorded 60 s capture of tier 1.

Considered and rejected for this demo:

- **Gephi Lite:** no timeline, only a date-range filter (verified docs).
- **Desktop Gephi:** not embeddable.
- **Neo4j NVL:** licence restricted to Neo4j products.
- **Memgraph Lab:** default render cap 3,500 nodes / 8,500 edges.
- **Kepler.gl LatAm arcs:** optional extra.
- **"Graphiphy":** no such viz tool. Graphify is a codebase-to-knowledge-graph skill for coding agents (useful to our agents, not for this graph).

11.2 Graph DB

- **None in the critical path.**
- Optional: Memgraph (Docker on the box) + Lab Community for a 2-minute Cypher Q&A on the pruned graph.
- Fallback: AuraDB Free (200k nodes / 400k relationships, verified docs). Too small for the full 90-day graph.
- Kuzu is archived.

11.3 TGNN (gated stretch; details: `graph_viz_tgnn_gcam.md` §4)

- **Library: TGM** (`tgm-lib`, MIT, beta, arXiv:2510.07586; EdgeBank, GraphMixer, TGAT, TGN, DyGFormer, TPNet, PopTrack) with **TGB 2.3.0** evaluators and negative samplers. DyGLib has been unmaintained since 2023.

- **Tasks:**

    - (A) entity→topic future links, scored with MRR on **new** edges, using historical + random negatives;
    - (B) next-week node traction, scored with NDCG@10 against LightGBM + SHAP.

- **Data:** 90 days ≈ 2–4M pruned events, with tone/arousal/fear/hope as event features. Chronological 70/15/15 split.
- **Compute (estimate):**
  - GraphMixer on a Modal L4: 0.5–1.5 h.
  - TGN on an A10: 1–3 h.
  - DyGFormer: 2–5 h on an A10.
  - $2–8 per model.
- **Explainability:** top contributing past interactions by leave-one-out re-scoring of the 20 most recent temporal neighbors, plus attention heatmaps. T-GNNExplainer has no maintained official code, so it is future work.
- **GO at 07:00 only if all hold:**
  1. New-edge MRR ≥ 1.10× the best of EdgeBank, PopTrack and persistence, over 3 seeds (> 2 sd).
  2. Reproducible and done by 07:00.
  3. Explanations work for the top 3 flagged edges.
  4. It doesn't hurt the traction NDCG.
- **Otherwise:** an honest baseline table slide.

12. Scoring model: six scores into the weekly top 10

#	Score	Inputs	Range
1	**Emergence**	z_7d, z_dow, growth_7d_log, accel_7d (GDELT shares) + new-neighbour rate + HN/X counts growth	z-like, then percentile
2	**Stage / saturation**	Onset order HN/arXiv → GDELT → Wikipedia; source diversity (n_sources); share of spa/por coverage; X creator diffusion	Seed / Emerging / Peaking / Saturated / Declining (Jev `choice` on the feature summary)
3	**Audience fit**	Jev `score` 0–4 on the topic summary; LatAm share; business/education subtopics	0–1
4	**Outperformance**	YouTube/X engagement vs. creator baseline for exemplars (not from GDELT)	Ratio
5	**Why it worked**	SHAP drivers from the LightGBM next-2-week traction model + driving graph edges + exemplar headlines	Text + evidence
6	**Emotional charge**	§10.4	z-like

Priority = `Emergence_pct × (1 − Saturation) × Fit × (1 + 0.25·EmotionalCharge_pct)`, ranked over topics and Leiden communities with ≥ 20 articles in the last 7 days.

- The 0.25 weight is a placeholder; tune it on the backtest.
- Each top-10 card shows the six chips, a sparkline, the framing angle, three evidence links, and "how 30X should adapt it" (format, hook, a 30–45 s outline).

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13. Evaluation

1. **Backtest:** rolling origin over the last 52 weeks, skipping the gap. - Label = topic's share growth over the next 2 weeks, cross-source where available. - Report **precision@10** and **lead time**: weeks between our flag and the news or Wikipedia peak. - Baselines: persistence, raw-count growth, degree growth, EdgeBank for new links.
2. **Filter quality:** precision per rule and language on 300 candidates; **recall on the control sample**; Wilson CIs.
3. **Editor eval:** a 30X-style editor (a team member, blind) rates the top 10 against a baseline top 10 (raw-count growth) for "would publish this week" (pairwise preference).
4. **Sentiment validation:** §10.7.

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14. Infrastructure and cost

Resource	Role	Cost (estimate unless verified)
BigQuery (user's project, on-demand)	Backfill, daily topic tables, rolling views, **GCAM 2 years**	$0 free tier + **~$12–20 on the ~$250 credit**
4 Claude Code accounts (tmux: claude-home, claude-work, claude-extend, claude-cacho)	WS1–WS4 + WS7 (TGNN) coding agents	Existing subscriptions
Gemini CLI	WS5 API + frontend (sigma.js story view, Graphistry iframe); Gemini on public YouTube URLs	Free quota
Grok CLI	WS6 entity resolution + cross-source + outperformance	Existing
Modal	Raw-file poller, API endpoint, **cuGraph layouts (L4 $0.80/h)**, **TGNN runs (A10 $1.10/h)** (verified prices)	~$10–20 for the weekend
Graphistry Hub	Tier-1 GPU graph view (server-GPU layout, iframe embed)	Free tier (resource limits; verified)
sigma.js / graphology / cosmos.gl	Tier-2 story view / tier-1 backup	MIT, $0
Nebius Token Factory	Bulk LLM labeling (Qwen3-30B $0.10/$0.30 per M; Qwen3-Embedding-8B $0.01/M, verified catalog). **UNVERIFIED that "Token Factory" = Nebius**	~$19 for 6 months
Featherless	Per-token Developer plan only (the Chat plan forbids automation, verified)	Optional
Hyperbolic	GPU rental only (H100 $3.19/h via tracker); serverless inference retired (third-party, UNVERIFIED)	Only if an interactive GPU box is needed
Jev (OpenRouter)	Typed decisions	~$1–2
X API	Counts for top-N topics	~$1–3
Lovable + Netlify	React UI, static JSON fallback	Free tiers

14.1 Budget table: the ~$250 Google Cloud credit plus other providers (estimates)

Line	Billed to	Estimate
Graph-ready backfill, daily_totals, aggregates	BigQuery free tier	$0–3

Line	Billed to	Estimate
GCAM ~108 days (demo)	GCP credit	$1.4–1.9
GCAM remaining 20 months	GCP credit	$8–12
GCAM/S9/S10 own-table reads, daily S6 + S7d	GCP credit	$1–3
Own-table storage (~20–40 GB)	GCP credit	~$0.5/month
GCP subtotal (hackathon)		**≈ $12–20 (≤ 8% of $250)**
cuGraph + TGNN	Modal	$5–15
LLM labels	Nebius (pending confirmation)	~$19
Jev + X counts	OpenRouter / X	~$2–5
Graphistry Hub, Netlify, Lovable, OSS	Free	$0
Weekend total		**≈ $40–60**

The remaining >$200 of credit covers months of daily operation, or a GCP GPU VM later. New projects often have a GPU quota of 0, so don't rely on that tonight (UNVERIFIED).

15. Multi-agent split and contracts

Contracts live in `contracts/`:

- `gkg_ai.schema.json`, `daily_topic.schema.json`, `topic_scores_weekly.schema.json`, `api.yaml`, `taxonomy.yaml`.
- Mocks in `mocks/` from minute one.
- Each agent owns one directory, works in its own git worktree on branch `ws<N>-…`, and updates `status/WS<N>.md` before each sync.
- Data never goes in git.

Key data contracts:

```
 radar.daily_topic (BigQuery) -> lake/gold/daily_topic.parquet
  dt, topic, lang_bucket, latam, n_articles, n_sources, share_of_day, mean_tone, sd_tone,
p10_tone, p90_tone,
  mean_polarity, extremity, positive_share, negative_share, controversy_index, is_missing (+
additive sums)
radar.daily_topic_rolling (view) -> lake/gold/daily_topic_rolling.parquet
  ... n_7d, share_7d, z_dow, z_7d, growth_7d_log, accel_7d, mean_tone_7d, extremity_7d,
controversy_7d, tone_shift_7d
lake/gold/node_metrics_weekly.parquet: week, entity_id, deg_w, deg_growth, new_nbr_rate,
pr_rank_delta, community_id, burst
lake/gold/topic_scores_weekly.parquet: week, topic_id, label_es, emergence, stage, audience_fit,
outperformance,
  emotional_charge, framing_angle, priority, rank, why JSON, evidence JSON
API: GET /api/radar?week=&lang=&limit=10 | /api/topic/{id}/graph | /api/topic/{id}/timeline |
/api/topic/{id}/sentiment | /api/freshness
```

15.1 Ready-to-paste briefs

Common preamble (paste first): *"Repo: radar. Work only in your directory on branch ws-... in your own git worktree. Read contracts/ first; never change it without a PR to the lead. Write outputs exactly as in the contracts. Add make test-ws validating outputs against the schema.*

Update status/WS.md before each sync (20:00, 22:00, 00:00, 07:00, 09:00, 10:30). No secrets in git. Mark every number as measured or estimated. Never zero-fill missing days: is_missing days are NULL."

WS0, human lead (with the merge captain):

- Run guardrails and Q1–Q4 (`bq_sql/`). Run S1 → S3 → S2 (newest month first) → S4 → S8.
- Commit contracts and mocks.
- Own the demo script and the decisions log.
- Approve any S7 GCAM run, after a dry run.

claude-work, WS1 BigQuery + ingestion (`pipelines/ingest/`):

1. Wrap `bq_sql/*.sql` in `run_bq.py`: dry run first, abort if more than 1.5× the ledger, log bytes per job to `reports/bq_ledger.csv`.
2. Run S2 for the remaining months, newest first, checking INFORMATION_SCHEMA.JOBS every 6 slices.
3. Export `gkg_ai`, `daily_totals`, `article_topic`, `daily_topic`, `daily_topic_rolling` and `edges_weekly` to Parquet (`export_own_tables.py`, free tabledata.list or extract jobs).
4. Port `gdelt_poc/poc_ingest.py` into a 15-minute https raw-file poller for today's partial day (Modal cron), parsing all 7 V2Tone fields.
5. Set up the S6 schedule (06:00 UTC); split it into 4 schedules if scripting is rejected.
6. DQ: `gold/dq_daily.parquet` with is_missing days.
7. **GCAM (S7):**
    - Dry-run October.
    - Raise the cap to 150 GiB for this job only.
    - Run Oct → Sep → Aug → Jul by 21:00, then the remaining months overnight, newest first.
    - Then run S9 and S10. Export `gcam_daily_topic`, `gcam_topic_rolling`, `viz_nodes_weekly` and `viz_edges_weekly`.
    - Schedule S7d daily after S6 (20 GiB cap).
    - Log bytes to `reports/bq_ledger.csv`. Restore the 50 GiB cap.

claude-extend, WS2 taxonomy, labels, quality (`pipelines/taxonomy/`):

1. Labeling functions with negatives (Italian "ai", Claude the person, gaming-only NVIDIA).
2. Nebius Qwen3-30B labels `{gkg_id, is_ai, subtopics[], latam, confidence}` for the last 8 weeks first, in batches of 20, resumable JSONL, with a cost log.
3. Qwen3-Embedding-8B + logistic regression routing.
4. Load the labels back into BigQuery (free load job) as `radar.article_label`; the topic tables use them when present.
5. Gold set: 300 candidates + 300 `f_control` → `reports/label_quality.md` with precision, **control-sample recall**, and Wilson CIs per language.

claude-home, WS3 graph (`graph/`):

1. Weekly igraph snapshots from `edges_weekly` (NPMI > 0.1).
2. Metrics: degree growth, new-neighbour rate, PageRank delta, k-core, Leiden, community matching (Jaccard ≥ 0.3), bursts.
3. Edge tone (mean tone of co-mentions).
4. Per-topic ego-graph JSON per week for the replay.
5. Mask gap weeks.

6. **GPU tier-1 pipeline (Modal L4, RAPIDS cuGraph):**
    - Build the 90-day full graph (n ≥ 3).
    - Compute warm-started weekly ForceAtlas2 (`max_iter=100`, `pos_list` = previous week), Leiden and PageRank.
    - Write `viz_full_nodes.parquet` / `viz_full_edges.parquet` with x, y, week, first_seen, tone, arousal and community.
    - Upload to **Graphistry Hub** with PyGraphistry (colors: tone diverging, community categorical) and publish the iframe URL in `status/WS3.md`.
    - Also write `web/public/data/full_graph.arrow` for the cosmos.gl backup.
7. Story-view JSON per week from S10 (≤ 400 nodes / 2,500 edges), plus topic ego graphs (≤ 80 nodes).
8. Optional: `memgraph_load.cypher` for the Q&A moment only.

claude-cacho, WS4 scores, sentiment, models (`models/`):

1. Features from `daily_topic_rolling` (emergence, sentiment), WS3 metrics, and WS6 lead-lag.
2. The six scores as in §12, the Emotional charge as in §10.4, and the framing angle as in §10.5 (Jev `choice` check).
3. LightGBM + SHAP on next-2-week share growth, as a rolling-origin backtest skipping the gap; baselines persistence, raw growth, EdgeBank.
4. `reports/backtest.md` (precision@10, lead time) and `reports/sentiment_validation.md` (the §10.7 test with WS6's YouTube/X data).
5. Write `gold/topic_scores_weekly`.
6. Stretch: TGN vs. EdgeBank on Modal.

Gemini CLI, WS5 API + frontend (`api/`, `web/`):

1. FastAPI implementing `api.yaml` (mocks first), with DuckDB over Parquet, deployed on Modal; `export_static.py` → `web/public/data/*.json` for the Netlify fallback.
2. Lovable UI (Spanish): - Top-10 cards with six chips and the angle badge - A sentiment panel (tone band p10–p90, pos/neg tails, controversy sparkline) - Ego-graph replay with a week slider - Onset timeline - Freshness badge - Backtest and quality cards
    - A community-sensor flow (mock)
3. **Graph views:**
    - Tier 1 = the Graphistry iframe (URL from WS3), with a recorded-video fallback.
    - Tier 2 = a **sigma.js v3 + graphology** story view: week slider with tweening, tone/arousal fill, fear/hope ring, NPMI tooltips, community colors, new-edge pulse, and a click → evidence side panel.
    - A parallel chart of share_7d and Emotional charge with a "we flagged it here" marker.
    - Backup: react-force-graph-2d.
    - Optional: a cosmos.gl full-graph page from `full_graph.arrow`.
4. Netlify deploy. Test on the demo laptop by 09:00.

claude-extend, WS7 TGNN (from ~22:00, once labeling runs unattended; `models/tgnn/`):

1. Build the events table from `article_topic` × `article_entities` × `gcam_ai` for 90 days: entities with ≥ 20 docs; event features tone, polarity, arousal, fear_pk, hope_pk, lang, latam.
2. TGM (`pip install tgm-lib`, beta): run EdgeBank, PopTrack and persistence first (CPU), then GraphMixer (Modal L4), then TGN (A10). DyGFormer only if time allows.
3. Use the TGB 2.3.0 negative samplers (historical + random). Report MRR overall and on **new** edges, 3 seeds, mean ± sd.
4. Explanations: leave-one-out re-scoring of the 20 most recent temporal neighbors for the top 3 flagged edges.
5. Write `reports/tgnn.md` with the go/no-go table (§11.3) by **07:00**. Never block WS4.

Grok CLI, WS6 entities + cross-source + outperformance (`entity/`, `crosssource/`):

1. Resolve the top 2,000 entities to Wikidata QIDs (wbsearchentities es/en/pt + P31 check); use GEG MIDs if Q4 shows they are cheap.
2. Weekly HN Algolia, Wikipedia pageviews (es/pt/en) and X counts for the top 50 topics → `gold/leadlag_weekly`.
3. YouTube RSS creator baselines plus 20–40 exemplar videos → outperformance ratios. Hook and format features come from Gemini on public URLs.
4. Three showcase stories for the pitch, each with onsets, driving edges and the emotional angle.

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16. Timeline: Sat 18:30 → Sun 12:00 (UTC-5), revised at 18:35

Time	Milestone	Owners
18:30–19:30	Next-30-minutes checklist (§20). Guardrails + $25/$50 alerts; Q1–Q4; S1, S3; contracts and mocks; agents briefed; Graphistry Hub account created	WS0, all
19:30–21:00	S2 newest 3 months → S4 → S8. **S7 GCAM Oct → Jul** (~108 days, ~$2) → S9. Modal cuGraph image built. Lovable screens + sigma.js skeleton on mocks. Labeling on 8 weeks	WS1, WS3, WS5, WS2
21:00 sync	First real `daily_topic` + `gcam_topic_rolling` for ~3 months; first top-10 draft; first Graphistry upload (1 week)	All
21:00–23:00	S2 months 4–12; S10 viz export; cuGraph weekly layouts; Graphistry 90-day view live; story-view JSON; scores v1 with GCAM terms; HN/Wikipedia/X series	WS1, WS3, WS4, WS5, WS6
23:00, cut line 1	Fewer than 6 months loaded → cap history at 6 months. Graphistry not working → promote cosmos.gl to tier 1. Jev failing → rule-based fit	WS0
23:00–01:30	S2 months 13–24. Backtest v1. Gold-set labeling. **WS7 TGNN baselines + GraphMixer**. Story view on real data. Sentiment validation data	All
01:30–06:30	Rotating sleep. **S7 GCAM remaining 20 months** (quota permitting). TGN run on Modal. S6 + S7d scheduled	WS1, WS7 (unattended)
07:00, TGNN go/no-go + sync	Backtest, quality, validation, `tgnn.md` reviewed	All
07:00–10:00	Polish: Spanish copy, top-10 briefs, demo data frozen for 2026-W42; **demo-laptop GPU test by 09:00**	WS4, WS5, WS0
08:00, cut line 2	TGNN only if GO; Memgraph/Kepler only if stable; cosmos.gl only if tested on the laptop	WS0

Time	Milestone	Owners
10:00, FEATURE FREEZE	Netlify production deploy + static fallback + recorded tier-1 video; rehearse twice	All
10:30–11:30	Submission text and screenshots	WS0
11:30	Submit (30-minute buffer)	WS0

Must-haves, in order:

1. Top 10 with "why" and the angle
2. Normalized daily series with gaps
3. Backtest card
4. Filter P/R card
5. **Story-view graph with evidence**
6. **Tier-1 GPU view** (Graphistry, or cosmos.gl/video)
7. Community-sensor flow

Nice-to-haves:

- 2-year GCAM and history
- TGNN (if GO)
- Memgraph Q&A
- Kepler.gl arcs
- docembed

17. Demo script (5 minutes) and judge narrative

1. **Hook (20 s):** "Paid scrapers tell you what landed on LinkedIn last week, and they break its terms. **We tell you what's about to land on LinkedIn, and from which emotional angle to tell it.**"
2. **Top 10 for week 2026-W42, in Spanish (60 s):** each card shows the six chips. Click the #1 topic: sparkline, z-score and growth, stage "Emerging", fit 0.8, angle "Debate".
3. **Why (60 s):**
    - SHAP drivers.
    - A 10 s Graphistry sweep over 90 days of AI news (the GPU view).
    - The sigma.js story replay for the #1 topic: new edges pulse, a community is born, nodes turn high-arousal; click a node for evidence.
    - Onset timeline: HN → GDELT es/pt → Wikipedia es.
4. **Emotion (45 s):** tone band, both tails, a controversy spike, the tone shift before the volume burst. The "both extremes" thesis comes with our validation number, whatever it is.
5. **Adapt (45 s):** the Spanish brief with format, two hooks in the angle, a 30–45 s outline, and evidence links.
6. **Trust (45 s):** - Backtest: "flagged N weeks before the peak, precision@10 = x vs. y for the baseline". - Filter recall on the control sample. - $0 BigQuery ledger. - ToS-clean source matrix and the community sensor for LinkedIn.
7. **Close (15 s):** the roadmap and the weekly radar as a 30X product.

18. Risk register

Risk	Likelihood	Impact	Mitigation
BigQuery row-count gap (477k vs. 645k) is a real undercount	Med	Med	Q1; normalize by BigQuery's own totals; or take that stream from raw files
Byte overrun on BigQuery	Low	Med	Max bytes billed 50 GiB, 0.7 TiB/day quota, dry runs, ledger CSV
GCAM accidentally referenced	Low	High ($)	Only in S7; cap blocks it unless raised deliberately
Regex topics noisy	High	Med	LLM labels replace them; P/R card shows the real numbers
News tone ≠ social emotion	High	Med	Explicit caveat; validation test; frame as "news emotional climate"
Scheduled query rejects the script	Med	Low	Split into 4 schedules
Agents collide on code	Med	Med	Worktrees, exclusive directories, contracts, merge captain
LLM provider rate limits (Nebius UNVERIFIED)	Med	Med	Embedding-classifier routing; label 8 weeks first
Backtest shows weak lead time	Med	High	Report honestly; emphasize normalized emergence and stage; show baseline comparison
Demo infra fails	Low	High	Static JSON fallback on Netlify; recorded video
ToS/privacy challenge from judges	Low	High	Source matrix; topic-level aggregates; no LinkedIn automation; consent notice in the sensor
GCAM cost overrun / credit not covering BigQuery	Low	Med	Dry-run each month; 150 GiB cap per job; $25/$50 alerts; check credit scope; demo needs only ~108 days
Graphistry Hub limits or network failure at the demo	Med	Med	cosmos.gl backup from precomputed coordinates; recorded video
Demo laptop GPU too weak (integrated GPU, hardware acceleration off)	Med	High	Test by 09:00; cap edges; story view is light (≤ 2.5k edges); fallback video
TGNN doesn't beat baselines or isn't ready	High	Low	Gated go/no-go; baseline table slide is the deliverable
TGM beta breaks	Med	Low	PyG TGN example / DyGLib fallback; or skip

19. Post-hackathon roadmap

1. **Weeks 1–2:** - LLM labels for 2 years. - Daily S6 in production. - Validation of the sentiment thesis on 300+ videos. - Instagram Graph API App Review.
2. **Month 1:** - Community sensor live with consent. - 30X own analytics as outcome labels. - Weekly digest to Slack/WhatsApp.
3. **Months 2–3:** - Wikidata-linked cross-source graph. - Chronos-2 residual anomalies. - TGN only if it beats EdgeBank on new links. - Optionally the 30X Voice model (SFT/DPO).

4. **Public asset:** an "AI in LatAm news" weekly page.

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20. Next 30 minutes (18:35–19:05, UTC-5)

1. **Guardrails:**
    - Max bytes billed 50 GiB (default).
    - Daily quota 0.7 TiB.
    - Budget alerts at **$1, $25 and $50**.
    - Check *Billing → Credits* to confirm the ~$250 credit covers BigQuery.
2. Run **Q1–Q4**, then **S1** and **S3**. Start **S2** for October 2026.
3. **Dry-run `s7_gcam_monthly.sql` for October.** If it's ≤ ~60 GB, set the cap to 150 GiB for this job, run it, and restore the cap.
4. Create a **Graphistry Hub** account (free) and a Modal secret for it. Run `pip install graphistry` in WS3's env.
5. Commit contracts and mocks. Paste the updated briefs (§15.1), including **WS7 TGNN** to claude-extend.
6. **Confirm:**
    - "Token Factory" = Nebius?
    - Jev access?
    - Merge captain?
    - Demo laptop model (integrated or discrete GPU)?

21. Decisions log

Decision	Status	Why
Early Signal Radar as the flagship; LinkedIn as *predicted uptake* + community signal	**Adopted**	Only ToS-clean way to answer "what's taking off on LinkedIn"
Opt-in LinkedIn browser extension	**Rejected**	LinkedIn bans "browser plugins and add-ons"; consent of the user doesn't override the platform terms
Paid scrapers (Apify/Bright Data)	**Rejected**	Violates the brief's no-ToS-violation rule
BigQuery single-pass monthly INSERT	**Adopted**	Two-stage costs ~25% more (0.71 vs. 0.57 TiB)
Two-stage extraction (IDs then join)	**Rejected** (except pilots / Storage Read)	Same reason
Wide net in BigQuery, precision locally + control sample	**Adopted**	Bytes don't depend on rows; recall becomes measurable
V2Tone fully parsed (7 fields)	**Adopted**	Zero extra bytes; enables the sentiment signals
Daily grain + weekday-adjusted rolling 7 days; weekly graphs	**Adopted**	Earlier signal than weekly; weekday seasonality handled
Unnormalized counts	**Rejected**	GDELT volume falls ~22%/year and varies by weekday; shares only
Zero-filling the 2025 gap	**Rejected**	Creates fake drops; NULL + mask

Decision	Status	Why
Sentiment as the sixth score + framing angle	**Adopted**	User's thesis; cheap (V2Tone); actionable for the writer
GCAM on 2 years	**Adopted** (revised 18:35)	With the ~$250 credit, 2 years ≈ 1.5–2.2 TiB ≈ $9–14; read once per month into a narrow table; demo window (~108 days) first
Two-tier GPU visualization (Graphistry wow + sigma.js story)	**Adopted**	GPU needed at 0.3–0.6M edges; explainability needs custom click/evidence
cosmos.gl + cuGraph-precomputed coordinates	**Backup**	In-browser GPU; demo-laptop risk
Gephi Lite / desktop Gephi as the demo viz	**Rejected**	No timeline in Lite (verified); desktop not embeddable
"Graphiphy"	**Not a viz tool**	Likely Graphify (a codebase knowledge-graph skill) or Gephi/Graphistry
Graph DB (Memgraph/AuraDB) in the critical path	**Rejected**; Memgraph optional for Q&A	Parquet → igraph/cuGraph → viz is simpler; AuraDB Free too small (200k/400k)
TGNN via TGM (TGN/GraphMixer)	**Gated stretch**	Go only if it beats EdgeBank/PopTrack/persistence on new-edge MRR by ≥10% (3 seeds) by 07:00
100-worker raw-file fan-out	**Rejected**	Impolite to GDELT servers; ≤16 workers suffices (the box PoC measured 2.7 files/s, 18.8 MB/s)
Graph DB for computation	**Rejected**	igraph/DuckDB/cuGraph are faster and simpler
TGNN as the core model	**Stretch**	Baselines (EdgeBank, persistence) often competitive; explainability worse than SHAP
gsg_docembed recall expansion	**Stretch**	Useful; only via Storage Read API
"Guaranteed permanent inflow" claim	**Rejected**	GDELT had a 17-day outage in 2025 and its APIs are in migration; we promise monitored freshness with gap detection instead
30X Voice model (post-training)	**Roadmap**	Too risky in 18 h
Jev for typed decisions	**Adopted**	Organizers' partner; calibrated, cheap

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22. Unverified and estimates

- **Unverified:**
  - Cause of the 477k-vs-645k BigQuery row gap (Q1).
  - **Whether the ~$250 credit covers BigQuery on-demand.**
  - Actual GCAM bytes per month (dry-run).
  - "Token Factory" = Nebius; Nebius rate limits; Jev limits.
  - Scheduled queries accepting the multi-statement S6/S7d.
  - Storage Read `row_restriction` with REGEXP/LIKE/_PARTITIONTIME.
  - Whether `gkg_partitioned` and `gsg_docembed` are clustered or partitioned; GEG size.
  - Graphistry Hub free-tier privacy, limits and embed timebar controls.
  - cosmos.gl v3 fixed-position parameters and React wrapper.
  - AuraDB Free including Bloom.
  - TGM beta stability and speed on our data.
  - Hyperbolic inference retirement and the H100 price (third-party).

    - GCAM derived-score redistribution terms.

- **Resolved since 18:05 (at 18:35):** Spanish ANEW fires on Spanish originals (99% of Spanish-source rows, measured; native-text processing verified in the GKG 2.1 codebook).
- **Hypotheses to test:**
    - Tone shift leads volume.
    - Emotional charge correlates with social outperformance.
    - High-arousal community birth precedes uptake.
- **Estimates:** all bytes, $, time, graph-size extrapolations (Heaps upper bounds) and TGNN/layout run times; score weights and GCAM thresholds are placeholders.

23. References (earlier reports in this folder)

- `report.md` / `.html` / `.pdf`: research report, alternatives, source matrix, Jev, costs, compliance, live checks.
- `gdelt_graph_analysis.md` / `.html` / `.pdf`: GDELT sizing, architecture, taxonomy, graph, TGNN, providers, 6-agent plan.
- `graph_viz_tgnn_gcam.md` / `.html` / `.pdf`: GCAM budget and dimensions, the two-tier GPU visualization comparison and storyboard, the graph DB choice, and the TGNN plan with go/no-go.
- `bq_free_tier_methodology.md` / `.html` / `.pdf`: BigQuery cost mechanics, per-column budget, SQL walkthrough, ledger, Run-these-now (Section 8b covers the Oct 17 update).
- `bq_sql/` and `bq_sql.zip`: q1–q4 probes; s1 setup (V2Tone columns, topic_rules); s2 monthly insert; s3 daily totals (lang_bucket, is_missing); s4 aggregates (entities, NPMI edges, article_topic, daily_topic); s5 export; s6 daily script; s7 GCAM monthly + s7d daily (46-dim narrow table + all-news baseline); s8 rolling view; s9 GCAM topic table + rolling view; s10 pruned emotion-annotated weekly graph for the viz; Python export and Storage Read tests.
- `gdelt_poc/`: measured PoC data (sizing, column sizes, tone calibration sample).
- `src/gkg21.txt`, `src/gcam.txt`: GKG 2.1 and GCAM codebooks used for verification.